American International University- Bangladesh (AIUB)
Faculty of Engineering (EEE)

Course Name :	Microprocessor and Embedded Systems	**Course Code :**	COE3104
Semester :	Summer 2024-25	**Sec :**	K
Lab Instructor :	MD. ALI NOOR		

Experiment No :	01
Experiment Name :	Familiarization with a microcontroller, the study of blink test and implementation of a traffic control system using microcontrollers.

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Performance Date:	16-7-25	**Due Date:**	08-08-25

Marking Rubrics (to be filled by Lab Instructor)

Category	Proficient [6]	Good [4]	Acceptable [2]	Unacceptable [1]	Secured Marks
Theoretical Background, Methods & procedures sections	All information, measures and variables are provided and explained.	All Information provided that is sufficient, but more explanation is needed.	Most information correct, but some information may be missing or inaccurate.	Much information missing and/or inaccurate.	
Results	All of the criteria are met; results are described clearly and accurately;	Most criteria are met, but there may be some lack of clarity and/or incorrect information.	Experimental results don’t match exactly with the theoretical values and/or analysis is unclear.	Experimental results are missing or incorrect;	
Discussion	Demonstrates thorough and sophisticated understanding. Conclusions drawn are appropriate for analyses;	Hypotheses are clearly stated, but some concluding statements not supported by data or data not well integrated.	Some hypotheses missing or misstated; conclusions not supported by data.	Conclusions don’t match hypotheses, not supported by data; no integration of data from different sources.	
General formatting	Title page, placement of figures and figure captions, and other formatting issues all correct.	Minor errors in formatting.	Major errors and/or missing information.	Not proper style in text.	
Writing & organization	Writing is strong and easy to understand; ideas are fully elaborated and connected; effective transitions between sentences; no typographic, spelling, or grammatical errors.	Writing is clear and easy to understand; ideas are connected; effective transitions between sentences; minor typographic, spelling, or grammatical errors.	Most of the required criteria are met, but some lack of clarity, typographic, spelling, or grammatical errors are present.	Very unclear, many errors.	
Comments:				Total Marks (Out of):	

Title of the Experiment:
Familiarization with a microcontroller, the study of blink test and implementation of a traffic control system using microcontrollers.

Introduction:
The Arduino Uno is a popular microcontroller board used for building digital devices and interactive objects that can sense and control physical devices. It features an ATmega328P microcontroller, 14 digital I/O pins, 6 analog inputs and a USB connection.

The objectives of this experiment are to-
1. Familiarize with the Arduino microcontroller
2. Implement a simple circuit to make an LED blink using the delay function
3. Implement a simple traffic control system

Theory and Methodology:
Arduino is an open-source platform used for creating interactive electronics projects. Arduino consists of both a programmable microcontroller and a piece of software, or IDE (Integrated Development Environment) that runs on your computer, used to write and upload computer code to the microcontroller board. Arduino Uno also doesn't need a hardware circuit (programmer/ burner) to load a new code into the board. We can easily load a code into the board just using a USB cable and the Arduino IDE (which uses an easier version of C++ to write code).

Overview of Arduino Board:

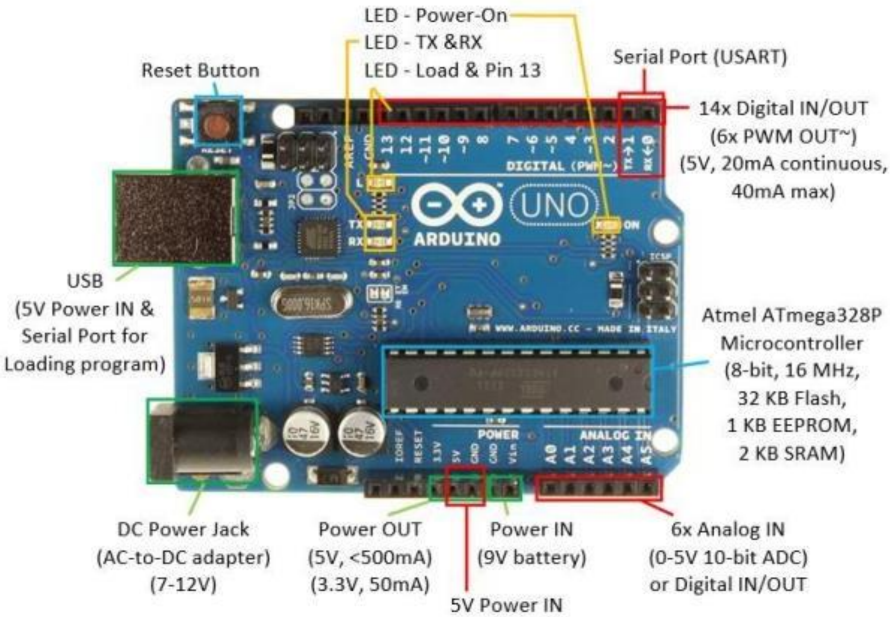

Figure-1: Arduino Uno Board [1]

Apparatus:
1. Arduino Uno (R3 board) or Arduino Mega 2560 board
2. Arduino IDE (Any version)
3. Breadboard
4. LED lights (red, yellow, green)
5. Three 220 ohms resistors
6. Jumper wires

Experimental Procedure:

A. Initial Setup

1. A USB cable Type-A to Type-B was used to connect the Arduino Uno board to the PC.

2. Arduino IDE was used to compile and push user code to the Arduino Uno board.

B. Using Arduino IDE to write code for a simple blink program [2]

1. The positive terminal of an LED was connected to Pin-13 of the Arduino Uno board, and the negative terminal of the LED was connected to the GND pin alongside a 220-ohm resistor which was set up on a breadboard.

2. The Arduino IDE was opened and on a blank sketch the following code was written-

```
void setup() {
  // put your setup code here, to run once:
  pinMode(13, OUTPUT); //Pin-13 selected as OUTPUT
}

void loop() {
  // put your main code here, to run repeatedly:
  digitalWrite(13, HIGH); //Turn LED ON
  delay(1000);            //LED ON for 1 second
  digitalWrite(13, LOW);  //TURN LED OFF
  delay(1000);            //LED OFF for 1 second
}
```

Figure-2: User code for blinking LED

The above code blinks the LED connected to Pin-13 of the Arduino Uno board, alternating between 1 second ON and 1 second OFF.

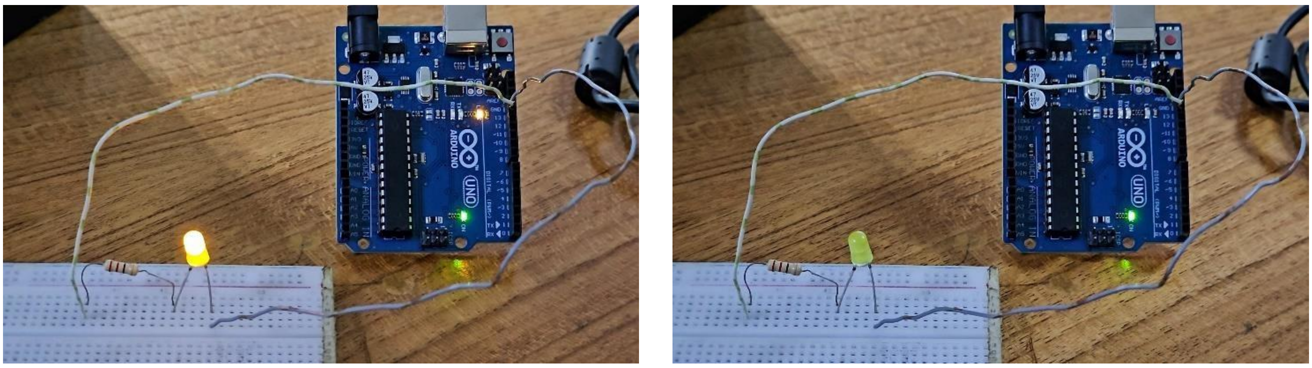

Figure-3: Hardware setup of Arduino Uno for blink test

C. Using Arduino IDE to write code for a traffic light control program [3]

1. On the breadboard, the positive terminals of three LEDs (Red, Yellow, and Green) were set up, each with a 220-ohm resistor at the negative terminal. Connections were then made with the Arduino Uno board as follows:

    - The positive terminal of Red LED was connected to Pin-13.
    - The positive terminal of Yellow LED was connected to Pin-12.
    - The positive terminal of Greem LED was connected to Pin-11.
    - All resistors at the negative LED terminals were connected to the GND pin.

2. The Arduino IDE was opened and on a blank sketch the following code was written-

```
void setup() {
  // put your setup code here, to run once:
  pinMode(13, OUTPUT); //Pin-13 selected as OUTPUT
  pinMode(12, OUTPUT); //Pin-12 selected as OUTPUT
  pinMode(11, OUTPUT); //Pin-11 selected as OUTPUT
}

void loop() {
  // put your main code here, to run repeatedly:
  digitalWrite(13, HIGH); //Turn Red LED ON
  delay(5000);            //LED ON for 5 second
  digitalWrite(13, LOW);  //TURN Red LED OFF

  digitalWrite(12, HIGH); //Turn Yellow LED ON
  delay(3000);            //LED ON for 3 second
  digitalWrite(12, LOW);  //TURN Yellow LED OFF

  for(int i=0; i<3; i++)  //Loop for 3 second blinking of Yellow LED
  {
    digitalWrite(12, HIGH); //Turn Yellow LED ON
    delay(500);             //LED ON for 0.5 second
    digitalWrite(12, LOW);  //TURN Yellow LED OFF
    delay(500);             //LED OFF for 0.5 second
  }

  digitalWrite(11, HIGH); //Turn Green LED ON
  delay(5000);            //LED ON for 5 second
  digitalWrite(11, LOW);  //TURN Green LED OFF
}
```

Figure-4: User code for Traffic Light Control System

The above code performs the following operations-

- Red LED ON for 5 seconds
- Yellow LED ON for 3 seconds
- Yellow LED Blink for 3 seconds, alternating between 0.5 seconds ON and 0.5 seconds OFF
- Green LED ON for 5 seconds

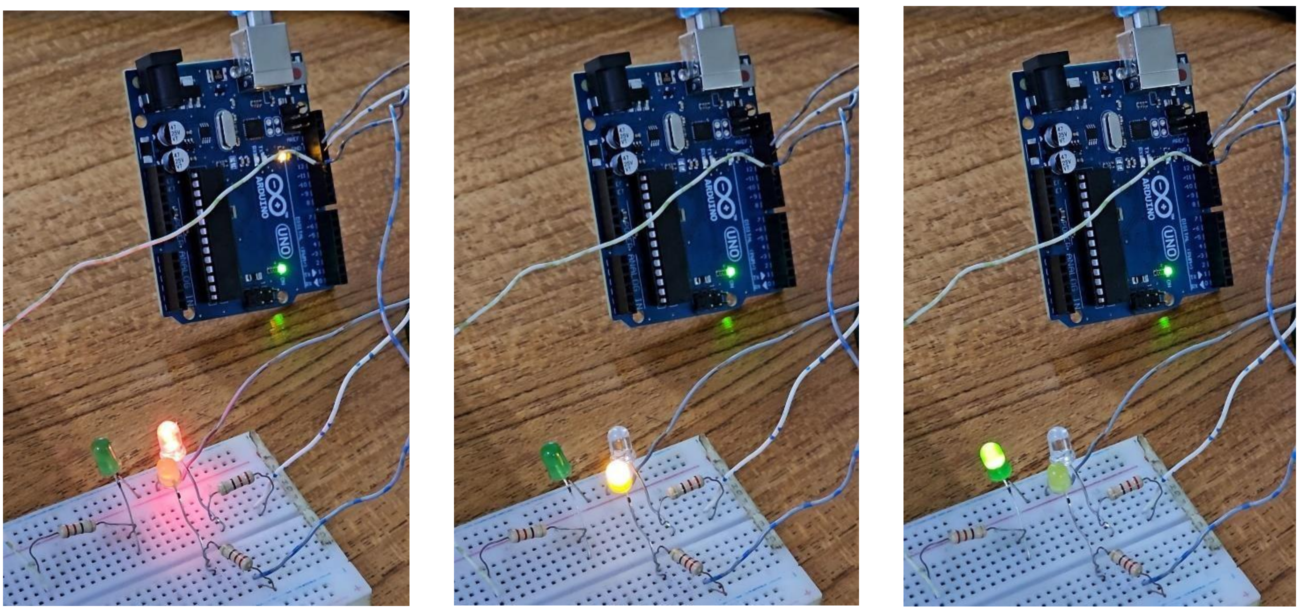

Figure-5: Hardware setup of Arduino Uno for traffic light control system

D. Compile and upload user code from Arduino IDE to Arduino Uno board

1. After the program was written, the sketch had to be saved. This was done by going to File->Save As, giving a file name (LED_Blink / TrafficControlSystem), and selecting Save.

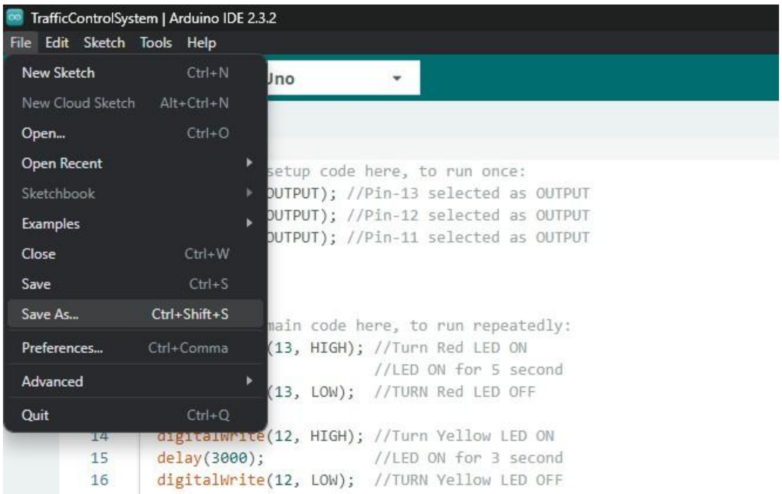

Figure-6: Save user code in Arduino IDE

2. Next, the code needed to be verified/compiled to identify and correct any errors. This was achieved by going to Sketch->Verify/Compile.

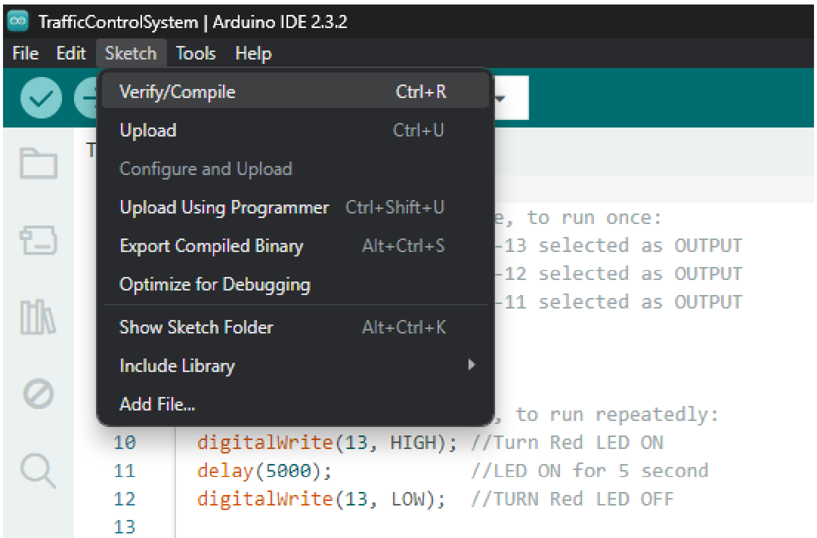

Figure-7: Verify/Compile user code in Arduino IDE

3. After the compilation was completed, the code needed to be uploaded to the Arduino Uno board. The Arduino Uno R3 board was connected to the PC using a USB cable.

4. Before uploading the code, the board type and port had to be selected in the Arduino IDE. This was done by navigating to:

   - Tools-> Board: -> Arduino AVR Boards -> Arduino Uno.
   - Tools->Port-> COMx.

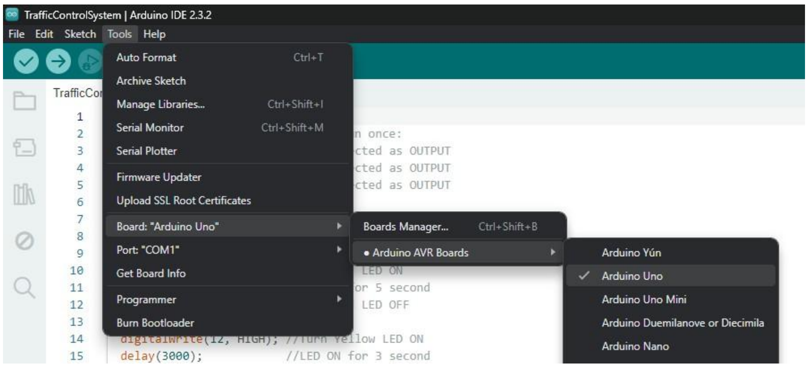

Figure-8: Select board type in Arduino IDE

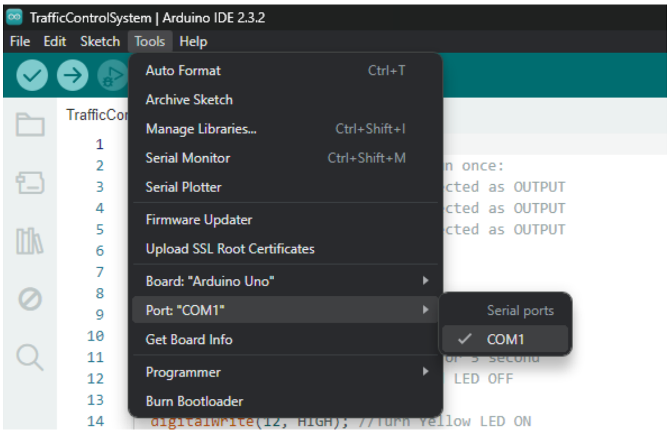

Figure-9: Select port in Arduino IDE to upload user code

5. After selecting the board and port, the upload option in the Arduino IDE was selected to upload the code to the Arduino board.

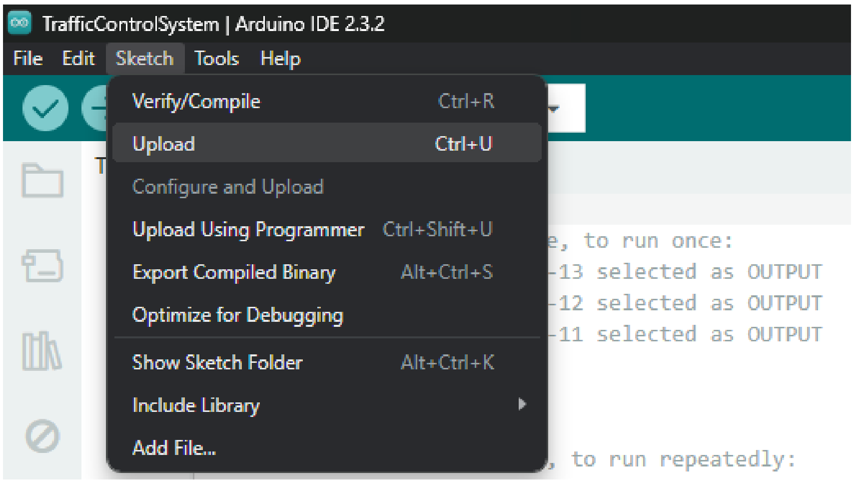

Figure-10: Upload user code from Arduino IDE

Simulation:

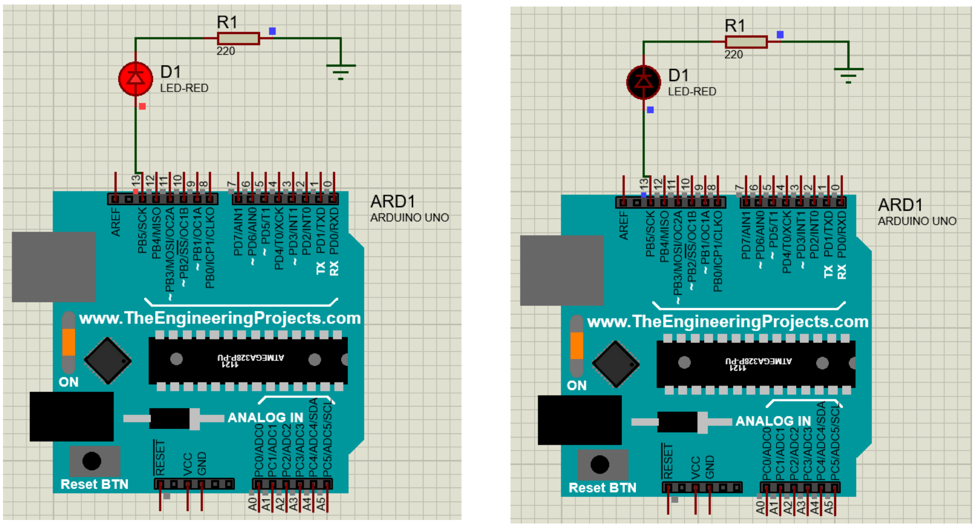

Figure-11: Simulation of blinking LED using Arduino Uno

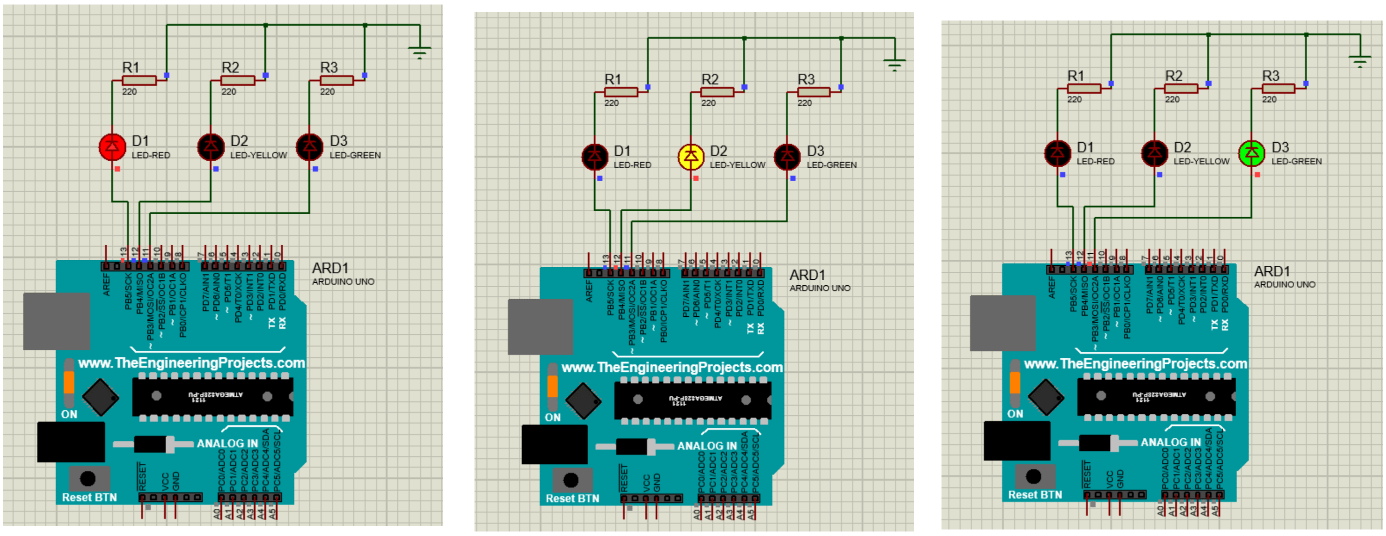

Figure-12: Simulation of traffic light control system using Arduino Uno

Conclusion & Discussion:

The Arduino microcontroller and the Arduino IDE were understood through the experiment. This understanding was crucial for future Arduino projects. In this experiment-

- The LED blink test was successfully performed, demonstrating the ability to control the digital I/O pins on the Arduino Uno board.

- A traffic light control system was implemented, introducing more complex programming concepts, such as controlling multiple LEDs with timing and sequence.

- Practical skills in building circuits were improved by working with the breadboard and components. The correct use of resistors to protect LEDs was practiced.

- The process of writing, verifying, and uploading code with the Arduino IDE was clearly demonstrated. This workflow was essential for successfully running any Arduino program.

- The simulations of both the LED blink and the traffic light control system were performed using Proteus software, confirming that the code worked as expected. The accuracy of the timing and logic in the programs was validated through these simulations.

Overall, the experiment was successful, achieving all objectives and demonstrating the effective use of Arduino for controlling electronic circuits.

Reference(s):

[1] "Arduino UNO R3." Available: https://docs.arduino.cc/resources/datasheets/A000066-datasheet.pdf. [Accessed: 07-Jun-2024].

[2] *Arduino.cc*. Available: https://docs.arduino.cc/built-in-examples/basics/Blink/. [Accessed: 07-Jun-2024].

[3] R. B. Devi, D. K. Reddy, E. Sravani, G. Srujan, S. Shankar, and S. Chakrabartty, "Density based traffic signal system using Arduino Uno," in *2017 International Conference on Inventive Computing and Informatics (ICICI)*, 2017, pp. 426–429.